

Pericyclic Reactions

Organic Chemistry II

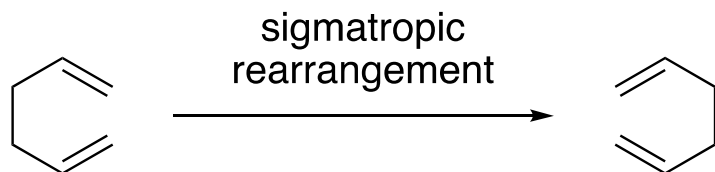
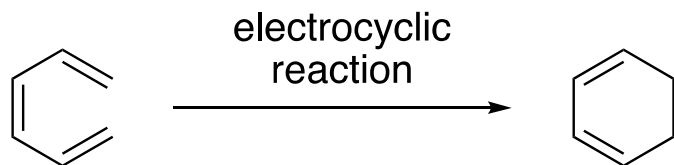
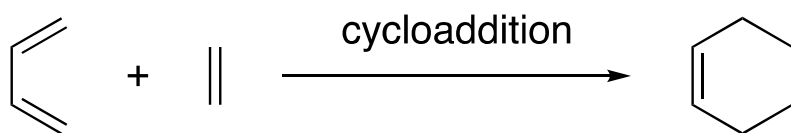
CHEM 2312

Dr. Michael Evans

1. Classify a pericyclic reaction as cycloaddition, electrocyclic reaction, or sigmatropic rearrangement; label a pericyclic process with appropriate notation.
2. Draw curved arrows for a pericyclic elementary step given the reactants and products.
3. Predict the products of Diels-Alder cycloaddition (4 + 2) reactions, including proper regiochemistry and stereochemistry.
4. Apply the Alder *endo* rule in predicting the products of Diels-Alder reactions.
5. Given the structure of a Diels-Alder adduct, draw the diene and dienophile that could be used to make it (i.e., apply the Diels-Alder reaction retrosynthetically).
6. Predict whether a given cycloaddition reaction under thermal or photochemical conditions is allowed or forbidden based on orbital symmetry.
7. Predict whether a given electrocyclic reaction is conrotatory or disrotatory based on orbital symmetry.
8. Predict the products of the Cope and Claisen sigmatropic rearrangements.

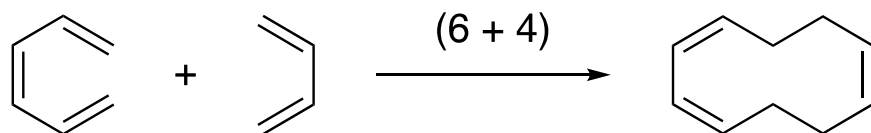
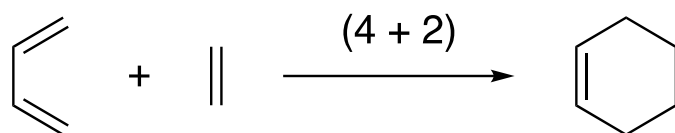
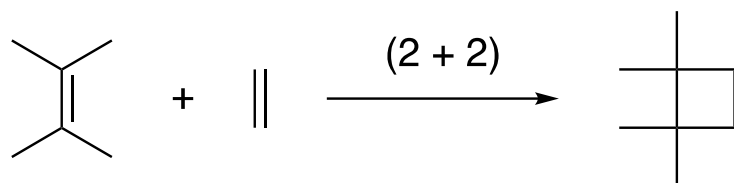
16.6 Pericyclic Reactions Involve Cyclic Electron Flow

- **Pericyclic reactions** do not involve radical or ionic intermediates (in fact, no intermediates at all)
- Electron flow is cyclic; reaction occurs through a single elementary step via a cyclic transition state
- Solvent polarity has little effect on reaction rate
- Three classes of reactions: **cycloadditions**, **electrocyclic reactions**, and **sigmatropic rearrangements**



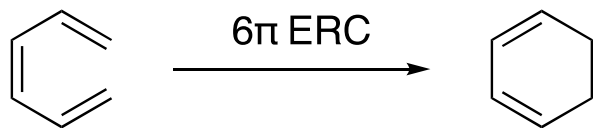
16.6 Cycloadditions Exchange Pi Bonds for Sigma Bonds

- **Cycloadditions** involve the union of two distinct pi systems
- Two sigma bonds are made and two pi bonds are broken (net)
- Named based on the sizes of the pi systems involved



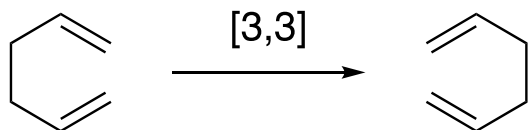
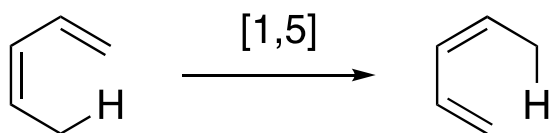
16.6 Electrocyclic Reactions Exchange a Pi Bond for a Sigma Bond

- **Electrocyclic reactions** involve the cyclization of a pi system (or the reverse, ring opening)
- One pi bond is broken and one sigma bond is made (net)
- Named based on the number of pi electrons



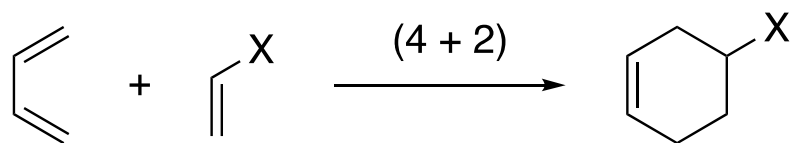
16.6 Sigmatropic Rearrangements Shift a Sigma Bond

- **Sigmatropic rearrangements** involve the movement of a sigma bond across a pi system
- Named based on the lengths of the pi systems on either side of the migrating sigma bond

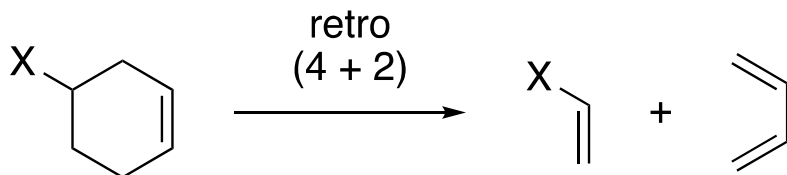


16.7 The Diels-Alder Reaction

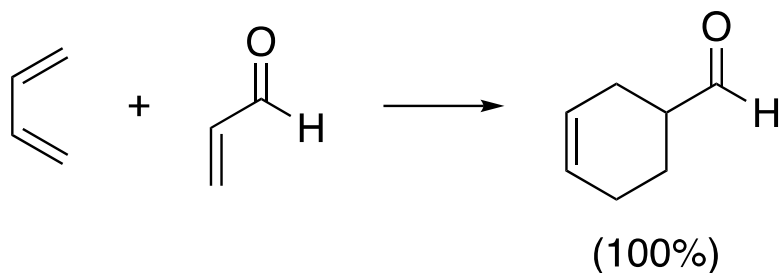
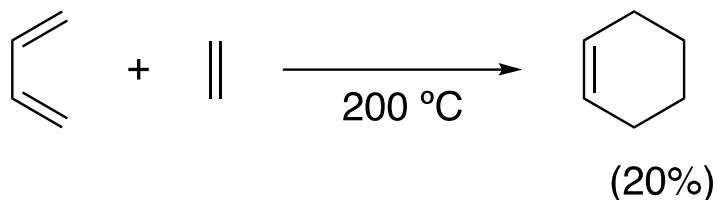
- (4 + 2) Cycloaddition is called the **Diels-Alder reaction**
- Fantastic synthetic utility...cyclohexenes and more
- Typically a concerted pericyclic mechanism although stepwise examples are known



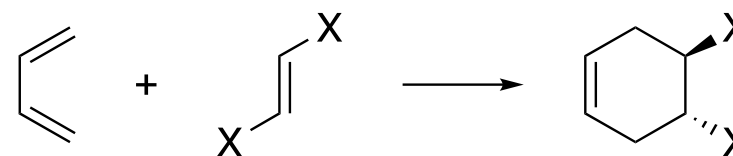
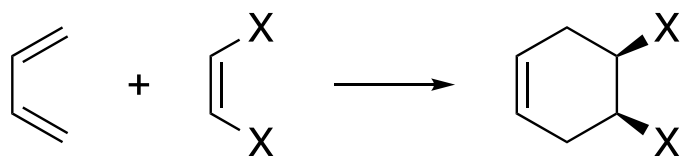
- High temperatures can promote **retro Diels-Alder reaction**



- Substituents on the dienophile lower the activation energy

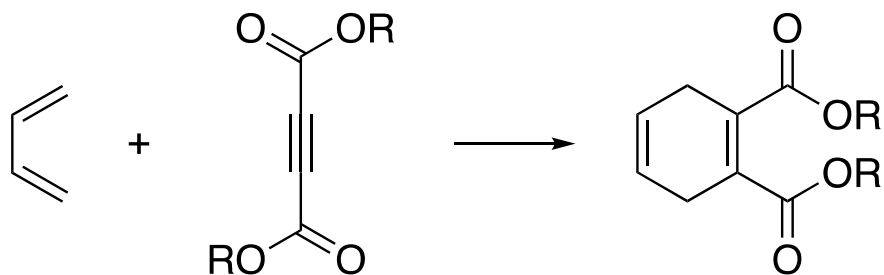


- Reaction is **stereospecific** with respect to the configuration of the dienophile: cis $\rightarrow$ cis, trans $\rightarrow$ trans

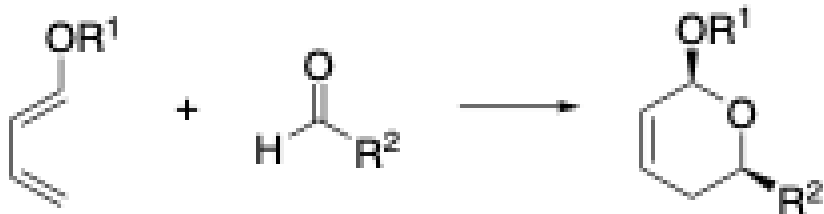


- Watch out for chiral products; remember to write + **En**

- The dienophile may also be an alkyne...

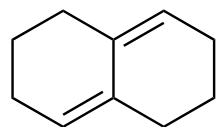
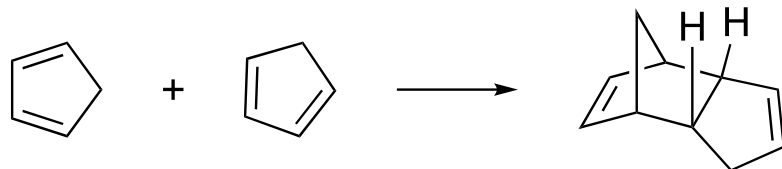


- ...or a double or triple bond involving a heteroatom!

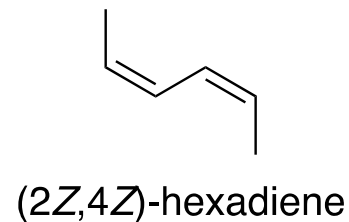
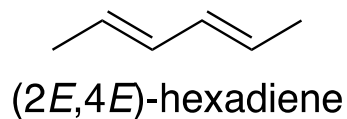


16.7 The Diene

- The reactive conformation of the diene is **s-cis**
- “Locked” dienes react either not at all (s-trans) or extremely rapidly (s-cis)

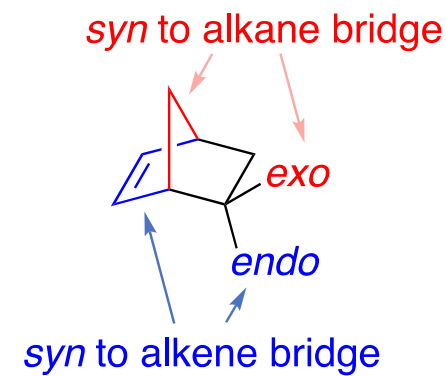
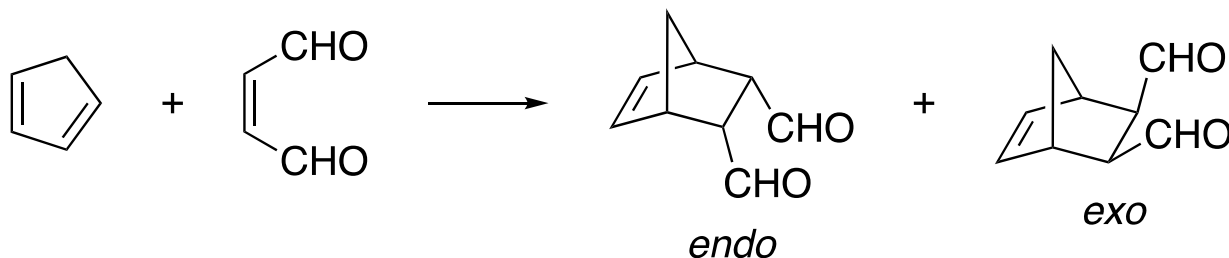


- Which of the following reacts more rapidly in (4 + 2) cycloaddition?



16.7 Substituents on the Dienophile may Approach *endo* or *exo*

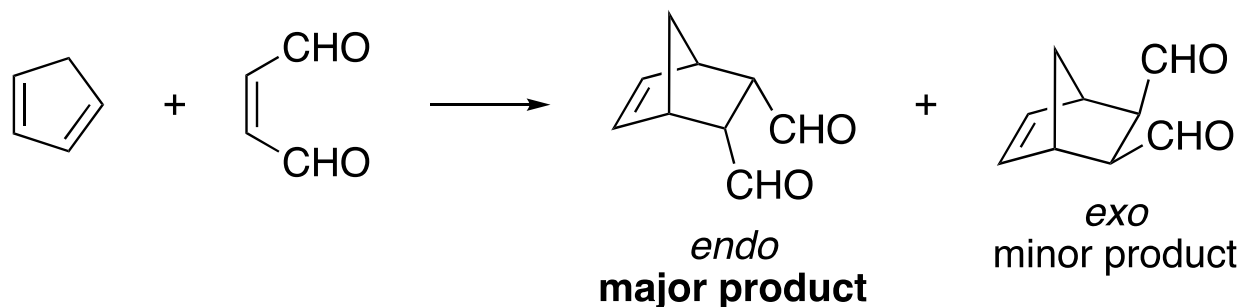
- Consider cyclohexene as a diene; dienophile substituents may end up *syn* or *anti* to the alkene (larger) bridge
- Substituents *syn* to the larger bridge approached the diene in an *endo* orientation
- Substituents *anti* to the larger bridge approached the diene in an *exo* orientation



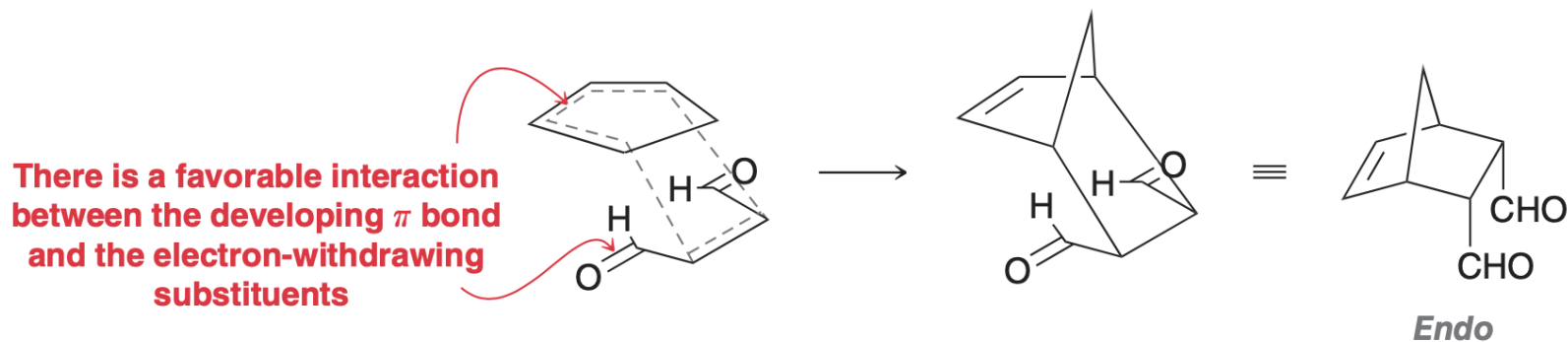
- Analyze the transition states leading to each product to confirm these stereochemical relationships

16.7 The Alder *endo* Rule

- The **Alder *endo* rule** states that the *endo* product is often favored in Diels-Alder reactions
- Particularly when the dienophile substituents contain pi bonds or lone pairs

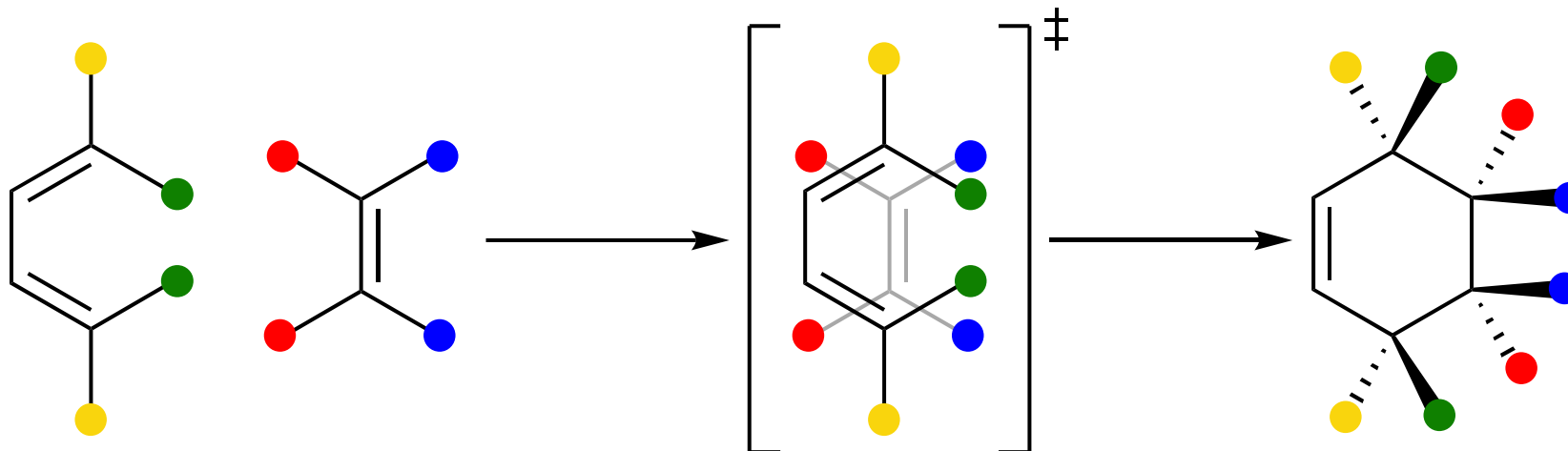


- Rationalized by stabilizing $\pi \rightarrow \pi^*$ or $n \rightarrow \pi^*$ orbital interactions between the dienophile substituents and the diene (**secondary orbital interactions**)



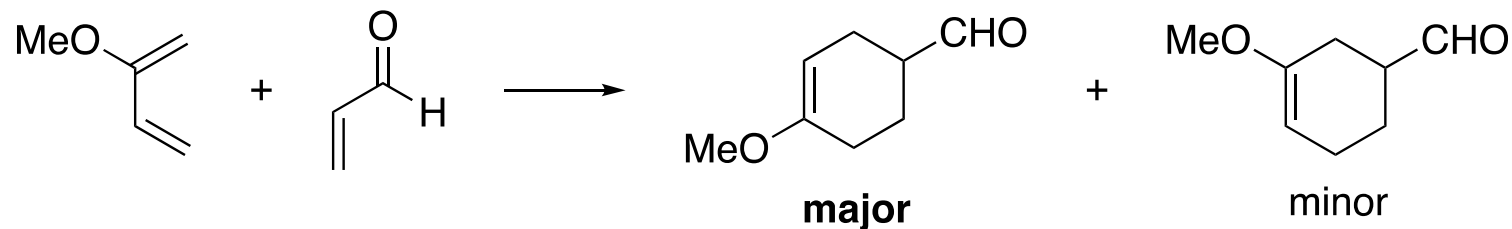
16.7 A General Transition State for Diels-Alder Reactions

- A general “scaffold” for the transition state of the Diels-Alder reaction is helpful for thinking in three dimensions
- Remember the **out-endo-cis** toggle rule!
 - Out + endo = cis
 - Out + exo = trans
 - In + endo = trans
 - In + exo = cis



16.7 Regiochemistry of the Diels-Alder Reaction (Think Resonance)

- If the diene *or* dienophile is symmetrical, there is no regiochemical issue (only one isomer possible)
- Two possible constitutional isomers (**regioisomers**) if the diene and dienophile are unsymmetrical



- Look for donating and withdrawing groups in the reactants, engage resonance, and connect + with –

16.8 Frontier Molecular Orbital Analysis of the Diels-Alder Reaction



- For butadiene + ethylene, considering either HOMO(diene) + LUMO(dienophile) or HOMO(dienophile) + LUMO(diene) shows that the Diels-Alder reaction is **symmetry allowed**

- When the reactants are polarized, the major interaction is HOMO of the electron-rich reactant + LUMO of the electron-poor reactant (still symmetry allowed!)

16.8 (2 + 2) is Symmetry Forbidden Thermally but Allowed Photochemically



- Because of phase mismatch between the HOMO and LUMO of two ethylene molecules, (2 + 2) cycloaddition under thermal conditions (heat) is **symmetry forbidden**
- When one of the reactants is photo-excited, (2 + 2) becomes allowed

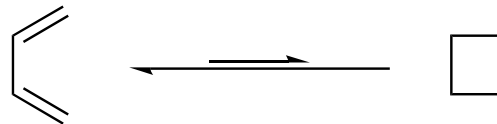
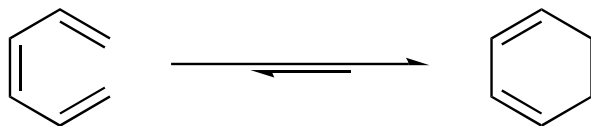
16.8 General Frontier Orbital Analysis of Cycloadditions



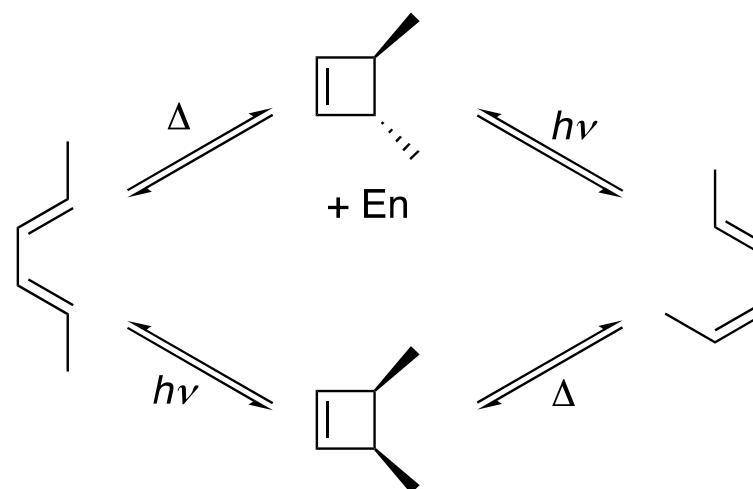
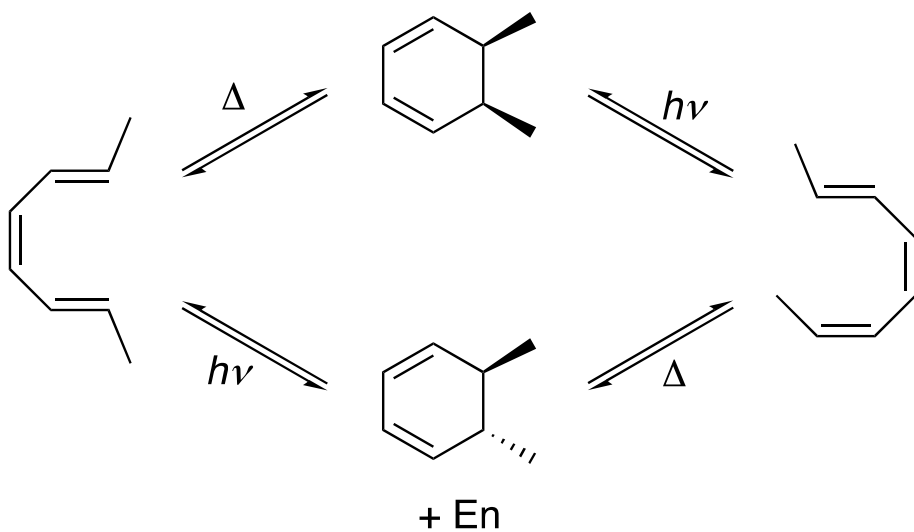
- Examine the HOMO and LUMO of the reactants...is simultaneous constructive overlap on both ends possible?
If so, the reaction is allowed; if not, forbidden

16.9 Electrocyclic Reactions

- Conjugated polyene undergoes cyclization via bond formation at the termini
- Position of equilibrium depends on ring strain



- When the termini bear substituents, there are distinct stereochemical outcomes under thermal and photochemical conditions *and* the length of the pi system plays a role

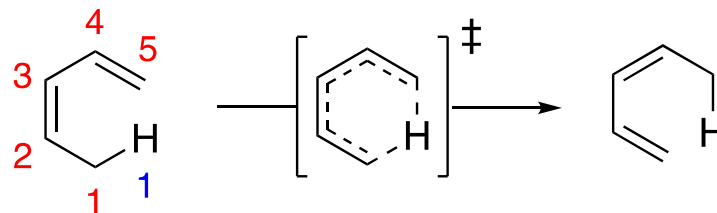


- For a four-atom conjugated system under thermal conditions, this occurs in a **conrotatory** manner

- For a four-atom conjugated system under photochemical conditions, electrocyclization is **disrotatory**

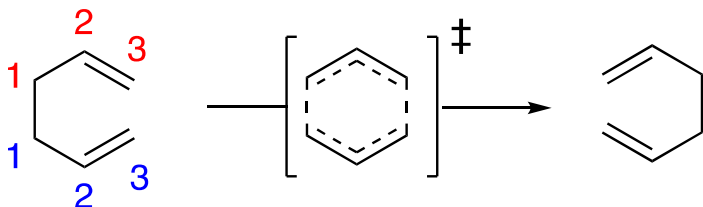
16.10 Sigmatropic Rearrangements

- A sigma bond migrates across one ($[1,n]$) or two ($[m,n]$) conjugated pi systems

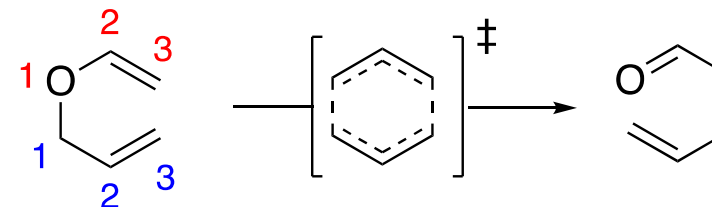


[1,5] sigmatropic rearrangement

- The [3,3]-rearrangement of a 1,5-diene is called the **Cope rearrangement**; the oxygen analogue involving allyl vinyl ethers is called the **Claisen rearrangement**



[3,3] sigmatropic rearrangement

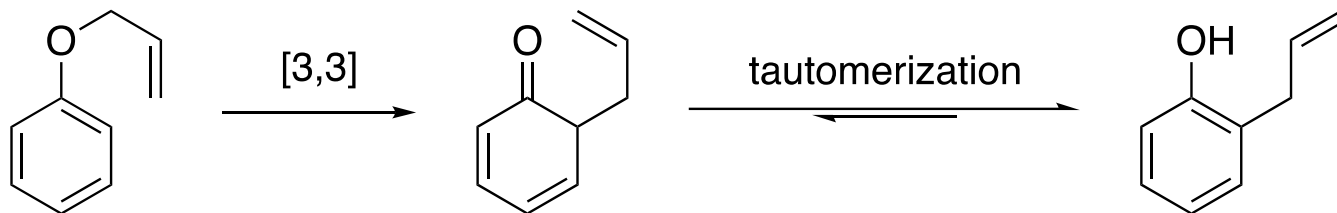


[3,3] sigmatropic rearrangement

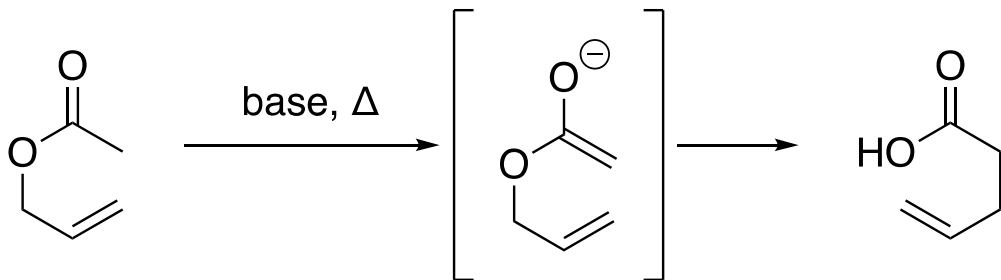
- These reactions favor the more substituted alkene (Cope) and the carbonyl group (Claisen)

16.10 “Hidden” Vinyl Ethers in the Claisen Rearrangement

- Allyl aryl ethers also participate in Claisen rearrangements



- Allylic esters engage in Claisen rearrangement via their enol forms (more on this later)



16.10 Anionic [2,3]: The Wittig Rearrangement

- [2,3]-Rearrangement of α -allyloxy carbanions is called the **Wittig rearrangement**
- Strong base deprotonates the starting alkyl allyl ether adjacent to the ether oxygen

